Study Number : **xxxxxxxxxxxxxxxxxx**

Study Title : Mini Ames Assay with Histidine Requiring Tester Strains of *Salmonella typhimurium*

Study status : Non-GLP

Sponsor : **xxxxxxxxxxxxxxxxxx**

Test Facility : Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad - 500 049, Telangana, India.

1.0 OBJECTIVE

The objective of this study was to assess the mutagenic potential of test compound **xxxxxxxxxxxxxxxxxx** using Mini Ames Assay.

2.0 MATERIALS AND METHODS

2.1 Test Item Information

Name of Test Item :

Batch No. :

Expiry/ Retest Date :

Solubility :

2.2 Vehicle Control Information

Water was used as a solvent for preparation of the test item formulation.

Name :

Batch No. :

Expiry Date :

Make :

2.3 Test System Information

The study was done using two tester strains of *Salmonella typhimurium* bacteria. The details of the test system are summarized in the below table.

In - Text Table: 01- Details of the Test System (Bacterial tester strains)

Bacteria	Strain	His Mutation	Other Mutations	Reversion Event
S. typhimurium	TA98	*his* D3052	*uvrB,rfa,* pKM101	frame-shift
S. typhimurium	TA100	*his* G46	*uvrB,rfa,* pKM101	base-pair

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Both tester strains were originally obtained from Molecular Toxicology, INC., Boone, NC, USA.

2.3.1 Genotyping: To ensure the genetic integrity of the test system, periodic genotype characterization was carried out as per test facility SOP and the results obtained were mentioned in the below table.

Last Genotype Performed Date: dd mm yyyy.

In - Text Table: 02- Genotype Characterization Details of Bacterial tester strains

Plate	Growth Characteristics	
	TA98	**TA100**
Minimal Agar Plates (auxotroph)		
Biotin Plates (His+ dependency)		
Histidine Biotin (His/Bio dependency)		
Histidine Biotin and Ampicillin (antibiotic resistance - plasmid PKM 101)		
Histidine Biotin and Tetracycline antibiotic resistance - plasmid PAQ1)		
Nutrient agar plates with Crystal Violet disc (*rfa* mutation)		
Nutrient agar plates Irradiation with Ultra violet light – Unexposed area (*UvrB* mutation)		
Nutrient agar plates Irradiation with Ultra violet light – Exposed area (*UvrB* mutation)		

2.4 Media and Solutions

All media and reagents required for conduct of the study were prepared and documented as per the applicable test facility SOPs.

2.5 Revival of Tester Strains

Frozen aliquots of bacterial cultures were thawed and cultures were inoculated using flask containing Oxoid Nutrient Broth (ONB) and incubated overnight in an incubator equipped with a shaker at 37±2 °C with shaking [approximately 110 revolutions per minute (rpm)]. Post incubation cell density **(**10^8 - 10^{10}) CFU/mL was ensured by OD measurement using UV-Spectrophotometer at 650 nm (Make: Shimadzu, Model: UV-2450).

These overnight grown cultures containing 10^8 - 10^{10} CFU/mL were used for the conduct of assay.

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2.6 Solubility Test

The test item formulation was prepared to achieve required stock. Solubility data was summarized as mentioned in the below table.

In - Text Table: 03 – Solubility Details of Test Item

Test Compound	Quantity	Solubility
XX		

2.7 Selection of Test Concentrations

Based on the solubility test results, XX/plate was selected as the highest concentration for treatment along with four additional lower concentrations each spaced by a factor of 2. Treatment method used was pre-incubation method in the presence of 10% metabolic activation system (For S9 mix details, please refer below **In-text table: 06**) and in the absence of metabolic activation system. Details of test concentrations and methods used were as mentioned in the below table.

In - Text Table: 04 – Concentrations of Test Item for respective Test Conditions

Test Concentration (µg /plate)	Treatment Method	Metabolic Activation System
XX	Pre-Incubation	Absent
		Present (10%)

2.8 Preparation of Test Item Formulations

Test item formulation was prepared freshly on the day of treatment. xx mg of test item was dissolved and made up to 3 mL using xxxxx by vortexing. The lower concentrations were prepared from its immediate higher concentrations using serial dilution with a spacing factor 2. The appearance of the prepared formulations was observed xxxxxxx solution.

2.9 Preparation of Positive Control Formulations

Preparation details of positive controls were summarized in the below mentioned table.

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In - Text Table: 05 – Details of Positive Control Formulation Preparation

Name	Strain	Test concentration	Source, Batch No.	Stock 1	Stock 2		
				Conc. (µg/ml)	Vol of stock 1 (µL)	Vol of solvent (µL)	Conc. (µg/ml)
2-AA	TA98 TA100	5 µg/plate					
2-NF	TA98	5 µg/plate					
SA	TA100	5 µg/plate					

Note: 2-AA: 2 -Aminoanthracene, 2-NF: 2-Nitrofluorene, SA: Sodium Azide.

Preparation of S9 Mix

The preparation of S9 mix details were mentioned in the below table.

In - Text Table: 06 – S9 Mix Details

S9 Fraction		Cofactor		S9 cofactor Mix (v/v)
Source, Batch no	Volume	Prepared On	Volume	

The prepared S9 mix was stored in refrigerator until further use in the study.

2.10 Labelling

All plates were suitably labelled to clearly identify the name of the tester strain, experiment method, test item concentration and vehicle control group.

2.11 EXPOSURE OF TEST ITEM

2.11.1 Pre - Incubation Method

In this method, 100 µL of vehicle control or test item or positive control, 0.5 mL of S9 mix/phosphate buffer and 100 µL of bacterial culture were mixed and incubated for minimum period of 20 minutes at 37 ± 2°C in a shaking water bath at 100 rpm. After incubation period, 2 mL of molten agar containing histidine-biotin solution maintained at 45 ± 3°C was added to this mixture followed by rapid mixing and pouring on to pre-labelled MGA plates. After solidification of agar, all treated plates were inverted and incubated at 37 ± 2°C in a BOD incubator (Make: xxxxx).

3.0 INCUBATION

All the treated plates were incubated at 37± 2°C for xx hours

4.0 RESULTS

4.1 Cytotoxicity and Precipitation

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Table: 01- Summary of Individual and Mean Revertants Data of Tester Strain TA98

Tester strain: TA98										
Type of Plate	**Treatment**	**Concn. (µg/well)**	**No. of Revertants**			**Mean**	**SD**	**BL**	**PS**	**FI**
			R1	**R2**	**R3**					
Absence of S9										
Untreated Control	NA	NA								-
Vehicle Control	xx									-
Test item	XX									
Positive Control	2-NF									
Presence of S9 (10%)										
Untreated Control	NA	NA								-
Vehicle Control	xx	0								-
Test item	XX									
Positive Control	2-AA									

Note: SD: Standard Deviation, BL: Background Lawn, PS: Precipitation score, FI: Fold Increase, Concn: Concentration, R1-R3: Replicate number, NA: Not applicable, N: No thinning, N*: mild thinning, NP: No Precipitation, 2-NF: 2-Nitroflourene, 2AA: 2 Aminoanthracene.

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Table: 02 - Summary of Individual and Mean Revertants Data of Tester Strain TA100

Tester strain: TA100										
Type of Plate	**Treatment**	**Concn. (µg/well)**	**No. of Revertants**			**Mean**	**SD**	**BL**	**PS**	**FI**
			R1	**R2**	**R3**					
Absence of S9										
Untreated Control	NA	NA								-
Vehicle Control	xx	0								-
Test item	XX									
Positive Control	SA					-	-			
Presence of S9 (10%)										
Untreated Control	NA	NA								-
Vehicle Control	xx	0								-
Test item	XX									
Positive Control	2-AA					-	-			

Note: SD: Standard Deviation, BL: Background Lawn, PS: Precipitation score, FI: Fold Increase, Concn: Concentration, R1-R3: Replicate number, NA: Not applicable, N: No thinning, N*: mild thinning, NP: No Precipitation, 2-NF: 2-Nitroflourene, 2AA: 2 Aminoanthracene.

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I, xxxxxxxxxxxxxxxxxx, hereby declares that the work was performed by myself and in accordance with the described procedures of study plan and standard operating procedures of APSL, Hyderabad.

Study Director

xxxxxxxxxxxxxxxxxxxxxxxx. _____

Signature

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Historical Control Data for Bacterial Reverse Mutation Test (Jun 2022)							
Tester Strain	**Treatment**	**Metabolic Activation System**	**Mean**	**±SD**	**N**	**Range**	
						Min	**Max**
TA 98	Organism	NA	36	14.43	208	10	81
	DMSO	-	37	13.62	180	12	70
	Water	-	41	13.38	49	18	73
	THF	-	24	5.47	12	16	36
	DMF	-	49	8.34	10	34	62
	Ethanol: water (1:1)	-	59	6.80	5	49	68
	PC: 2-NitroFluorene	-	532	202.33	206	222	1129
	DMSO	+	38	9.74	176	16	68
	Water	+	38	8.79	50	16	54
	THF	+	32	8.03	12	16	40
	DMF	+	36	7.14	15	22	48
	Ethanol: water (1:1)	+	47	5.03	5	39	52
	2 Amino anthracene	+	1394	461.21	201	102	3724
TA 100	Organism	NA	156	33.15	282	75	243
	DMSO	-	144	36.69	239	81	240
	Water	-	156	41.13	71	72	226
	THF	-	124	28.10	20	86	194
	DMF	-	185	22.54	15	132	215
	Ethanol: water (1:1)	-	185	19.47	9	158	226
	Positive Control: Sodium Azide	-	1506	387.02	261	716	2568
	DMSO	+	155	33.84	235	91	266
	Water	+	157	36.01	69	95	240
	THF	+	140	41.16	20	98	284
	DMF	+	175	16.16	15	136	197
	Ethanol: water (1:1)	+	203	24.67	9	169	235
	2 Amino anthracene	+	1445	490.04	259	432	2986

PC: Positive Control

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1.0 OBJECTIVE:

1.1. The objective of this Standard Operation Procedure (SOP) is to describe the procedure for the conduct of the bacterial reverse mutation Test (BRMT).

2.0 SCOPE:

2.1. This SOP is applicable for the procedures related conducting of the Bacterial Reverse Mutation Test (BRMT) at Toxicology Department, Aurigene Pharmaceutical Services Limited (APSL), Bollaram Road, Miyapur, Hyderabad, Telangana, India.

3.0 RESPONSIBILITY:

3.1. The Study Director/Study Personnel is responsible for implementation of this SOP.

3.2. QA Personnel Should be aware of this SOP.

3.3. Head Function-Toxicology and Test Facility Management are responsible to ensure the overall compliance to this SOP.

4.0 DEFINITION(S):

4.1. **Bacterial Reverse Mutation Test (BRMT):** BRMT is a rapid, reliable and economical method for screening test items for potential mutagenicity at the genetic level. It employs bacterial strains (tester strains) as the test system. The test detects the mutations which revert in the tester strains and restore the functional capability of the bacteria to synthesize essential amino acid Histidine/Tryptophan. The reverted bacteria grows auxotrophically in the absence of amino acid required by the parent tester strains.

4.2. **Tester Strains:** Tester Strains are a set of histidine-requiring strains of *Salmonella typhimurium* and tryptophan requiring strains of *Escherichia coli* used for mutagenicity testing. Each tester strain contains a different type of mutation in the histidine operon. Recommended strains for BRMT are *Salmonella typhimurium* TA1537, TA1535, TA98, TA100 and TA102 /or *Escherichia coli* (*E.coli*).

4.3. **Untreated Control:** The untreated control provides baseline data of the test system under the conditions of the assay**.** Untreated control shall be maintained when other

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than routine solvent is used whose composition will be Tester strain + Top agar plated on MGA plate.

4.4. **Negative Control:** The negative control provides evidence that the test system is not responsive under the actual conditions of the assay**.** Negative control shall be maintained when other than routine solvent is used where the mix shall contain Tester strain + S9/buffer + Top agar plated on MGA plate. .

4.5. **Vehicle Control:** The vehicle control provides evidence for a lack of influence of the chosen vehicle on the test system under the actual conditions of the assay. The solvent/vehicle in which test item is solubilized and formulated. Vehicle control mix shall have Tester strain + S9/buffer + Solvent + Top agar plated on MGA plate.

4.6. **Positive Control:** The positive control provides evidence that the test system is responsive under the actual conditions of the assay**.** Positive control mix shall have Tester strain + S9/buffer + Positive Control + Top agar plated on MGA plate.

5.0 ABBREVIATION(S):

Abbreviation	Full Description
APSL	Aurigene Pharmaceutical Services Limited
BSC	Bio Safety Cabinet
BRMT	Bacterial Reverse Mutation Test
DRF	Dose Range Finding Assay
GLP	Good Laboratory practices
H	Histidine
ID	Identification Number
LAFC	Laminar Air Flow Cabinet
MA	Mutagenicity assay
MGA plates	Minimal Glucose Agar plates
mL	Millilitre
NA	Not Applicable
No.	Number
OECD	Organization for Economic Co-operation and Development
PB	Phosphate Buffer
PC	Positive Control
PPE	Personal Protective Equipment
rpm	Rotations per minute
SOP	Standard Operating Procedure
TICO	Test Item Control Officer
Trp	Tryptophan

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6.0 JOB SAFETY ANALYSIS

S. No	Potential Hazards	Recommended Safe Job Procedure
1.	Physical injury – Handling solvents	Ensure to use appropriate PPE in handling solvent and solvent containers. Use secondary containers appropriately wherever applicable.
2.	Physical injury – Handling media and reagents	Ensure to use appropriate PPE while handling media and reagents. Use appropriate containers to hold media/reagents.
3.	Physical injury – Handling of glassware and broken glassware	Ensure to use appropriate PPE in handling glassware. Use secondary containers appropriately. Ensure to use appropriate PPE along with cut-resistant gloves while handling broken glassware and sharp-edged glass items. Disposal of broken glassware shall be disposed of in bags of respective color bins segregated as broken glassware hazardous.
4.	Short circuit and Electric Circuit	Check the electrical wires and socket integrity visually before switching on the equipment. Ensure preventive maintenance is performed as per the schedule for all equipment that are involved in the media/reagent preparation.
5.	Physical injury-Equipment handling	Ensure preventive maintenance is performed as per the schedule for all equipment that are involved in the media/reagent preparation. Use appropriate PPE while handling equipment's.
6.	Contamination of cells & media	Use appropriate PPE while handling contaminated broth, media, agar plates and bacterial cultures. Sterilize the materials and LAF work area (before and after use) with 70% alcohol or with any available microbial disinfectant. Use autoclaved materials where ever appropriate.

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7.0 CAUTIONS AND INTERFERENCES

7.1. All recommended and required safety precautions including proper use of PPEs must be used while handling tester strains, test item/positive control and during disposal.

7.2. Follow all aseptic procedures during conduct of the assay wherever applicable.

7.3. Plan the experiments/activities as mentioned in the study plan for the requirements like selection of tester strains, solvents, S9 fraction concentration, test item formulations, positive controls formulations, formulation analysis (if applicable) etc.

8.0 PROCEDURE:

8.1. Prepare all media, reagents, formulations and bacterial cultures etc. required to conduct BRMT by following respective SOP's:

8.1.1 **Media and Solutions:** Refer SOP**: SOP-GLP-TOX-0045** for media and reagents preparation.

8.1.2 **Tester Strains Handling and Culture viability measurement:** Refer SOP: **SOP-GLP-TOX-0046.**

8.1.3 **Positive Controls:** Refer SOP: **SOP-GLP-TOX-0044** for preparation of positive controls.

8.1.4 **Hazardous and Spillage Handling:** Follow procedures as per **SOP: SOP-GLP-SHE-0002, SOP-GLP-SHE-0003 and SOP-GLP-SHE-0004.**

8.1.5 Perform critical activities related to BRMT at sterile conditions in LAFC/BSC by following **SOP-GLP-GEN-0021**.

8.2 Test Item Solubility and Precipitation Check Test item for which either solubility data is not available or not provided by the sponsor or needs confirmation, check solubility of test item as a part of study or as feasibility check before study initiation to choose appropriate solvent.

8.2.2 Test item will be checked for its solubility in different vehicles in the preferential order starting with water (most preferred for this study type). If aqueous formulation is not a clear solution, DMSO or other recommended compatible vehicle will be tested until a clear solution is obtained.

8.2.3 Other vehicles include dimethyl formamide, acetonitrile, ethanol (95 %) and Tetrahydrofuran. If other than well-known vehicles are used, or not supported by in-

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house historical control data their inclusion in the study shall be supported by data indicating their compatibility with survival of test system, S9 activity and shall not involve in chemical reaction with test item. The highest test item concentration tested for solubility should be 50 mg/mL, if not soluble at 50 mg/mL lower concentrations should be tested.

8.2.4 The recommended maximum test concentration for soluble, non-cytotoxic substances is 5 mg/plate or 5 µL/plate. For non-cytotoxic substances that are not soluble at 5 mg/plate or 5 µL/plate, one or more concentrations tested should be insoluble in the final treatment mixture. The observed precipitate shall not interfere with the scoring.

8.2.5 If, precipitation test is performed before or during conduct of DRF phase, take 100 µL of highest concentration stock of test item and add to 2 mL of top agar. Mix gently and plate on to MGA plate. Allow the top agar to solidify and observe for any visible precipitate on the plate. Document the details and observations in format **FORM-GLP-TOX-0077.**

8.2.6 Observe precipitation of test item in the tested concentrations during treatment and record the details as Precipitation presence or absence in the format **FORM-GLP-TOX-0075.**

8.3 Tester Strains

8.3.1 Ensure availability of overnight grown bacterial cultures for conduct of experiment.

8.3.2 Measure the cell density of cultures as per **SOP-GLP-TOX-0046** either by determining OD or by viability count method. Record the details in **FORM-GLP-TOX-0067.**

8.3.3 Consider the cultures with densities ranging from 10^8 to 10^{10} cfu/ml, suitable for conduct of assay.

8.4 Preparation of Test Item

8.4.1 Obtain the test item from TICO for BRMT and Sponsor for mini AMES test.

8.4.2 Follow the **SOP-GLP-TC-0003** to obtain the test item request according to the study requirement.

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8.4.3 Follow the instructions provided by the sponsor/ as per the study plan to prepare the test item formulations.

8.4.4 Add liquid test item directly on to the MGA plate or dilute in the solvent/vehicle or consider as specified in the Study Plan.

8.4.5 Fresh formulation preparations should be employed unless stability data demonstrates the storage duration of formulations. Apply correction factor during preparation of test item formulations based on test item purity or potency on need basis.

8.4.6 Maintain a spacing factor among the concentrations prepared, not exceeding approximately 3.16 for the experiments and maintain a minimum of five analyzable concentrations for the experiment.

8.4.7 Prepare formulation of different concentrations by dissolving required quantity of test item in appropriate volume of vehicle or by making dilutions from stock.

8.4.8 An example illustrating the steps involved in the test item formulation preparation by diluting stocks is elaborated below;

8.4.9 For a 50 mg/ml stock preparation, weigh 250 mg of test item and dissolve it in 5 mL of selected vehicle and label it as Stock S1. Alternatively, adjust vehicle volume based on the amount of test item weighed.

8.4.10 Make 1:10 dilution of Stock S1 with vehicle to get Stock S3 (5 mg/mL).

8.4.11 Make 1:10 dilution of Stock S3 with vehicle to get Stock S5 (0.5 mg/mL).

8.4.12 Weigh 75 mg of the test item and dissolve it in 5 mL of suitable vehicle to get Stock S2 (15 mg/mL). Alternatively, adjust vehicle volume based on the amount of test item weighed.

8.4.13 Make 1:10 dilution of Stock S2 with the vehicle to get Stock S4 (1.5 mg/mL).

8.4.14 Alternatively, prepare lower dilutions by diluting the primary stock using the following formula;

$C_1V_1=C_2V_2$

Where, V_1 = Volume of primary stock to be taken

C_1 = Concentration of primary stock available

V_2 = Volume of working stock to be prepared

C_2 = Concentration of working stock to be prepared

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8.4.15 For test concentrations of 5000, 1500, 500, 150 and 50 µg/plate, use 100 µL of test item stocks of 50 (S1), 15 (S2), 5 (S3), 1.5 (S4) and 0.5 (S5) mg/mL, respectively for treatment.

8.4.16 Observe and document the remarks regarding the formulations like appearance/color on need basis or as per study plan specification.

8.4.17 Document details of preparation using format **FORM-GLP-TOX-0069.**

Note: *Provide aliquots of highest and lowest analyzable test concentration for dose formulation analysis, if the formulations are prepared by serial dilution method. Incase if formulations are prepared individually, submit all strengths for formulation analysis or as specified in the study plan.*

8.5 Preparation of Positive Controls

8.5.1 A list of positive controls recommended in OECD Test Guideline 471 are given in Work Instruction **WI-GLP-TOX-0029**.

8.5.2 Weigh appropriate amount of individual positive controls and dissolve them in the solvent preferably in water or DMSO as listed in the annexure.

8.5.3 Make appropriate dilutions so as to achieve the desired concentration per plate.

8.5.4 Alternatively, use vendor supplied readymade positive controls frozen stocks or master stocks and document the details. Follow handling of PCs as listed in the safety data sheets provided by the supplier.

8.5.5 Document details of positive control preparation using format **FORM-GLP-TOX-0070.**

8.6 Preparation of S9 mixture

8.6.1 The post-mitochondrial fraction (S9) can be used at concentrations in the range from 5 to 30% v/v in the S9 mixture.

8.6.2 As per study plan requirement, select the type and source of S9.

8.6.3 Eg. S9 Source: Sprague dawley, wistar rats, hamster etc.

8.6.4 S9 type: Induced (Phenobarbitone or Aroclor induced etc.) or Uninduced

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8.6.5 Prepare S9 mix when metabolic activation system is required/ as specified in the study plan. Calculate the total volume of the S9 mixture required for the assay and accordingly thaw appropriate volume of S9 and cofactor to prepare S9 mixture.

8.6.6 Mix appropriate volume of S9 to cofactor mix to obtain the desired percentage of S9 cofactor mix. Refer table below as an example for preparation of 5 and 10 % S9 cofactor mix.

Ingredients	Per 100 mL		
	10%	5%	Units
Cofactor for S9 mixture	90	95	mL
Rat liver S9	10	5	mL

8.6.7 Always prepare fresh S9 mix and maintain at ~ 2-8˚C by placing in refrigerator or on ice / ice packs until use.

8.6.8 Document the receipt and utilization of S9 using format **FORM-GLP-TOX-0076** and maintain the records in the facility.

8.6.9 Document details of S9 cofactor mix preparation using format **FORM-GLP-TOX-0075.**

8.7 Labeling

8.7.1 Label the plates prior or during the conduct of experiment with the minimum details to identify each plate. An example of labeling contents details are: Study no / test method/phase/ strain/ metabolic activation/ treatment group or code/ replicate/etc.

- Illustration:

G-22-001/PI- DRF
TA98/S9/ T2 R1

- G-22-001: Study number
- PI/PrI-Plate Incorporation/Pre-Incubation method
- Phase: DRF-Dose Range finding assay, MA-Mutagenicity Assay
- TA98 – Tester strain

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- (-S9/S9) – Absence/Presence of metabolic activation system
- T2- Test item treatment concentrations no 2.
- PC-Positive Control
- VC-Solvent/Vehicle control
- R1 – Replicate no 1.

8.7.2 During experiment, arrange the plates and containers/tubes used to mix the contents before pouring, in sequential order of treatment. For example, if plates are arranged in the order of VC (R1, R2, and R3), Test item concentrations and PC, then arrange the pouring containers in the same order.

8.8 Study Design

8.8.1 Design the study in two experiments viz. DRF and MA or as specified in Study Plan.

8.9 Dose range finding Experiment (DRF)

8.9.1 DRF is designed to assess the cytotoxicity with an objective to select highest test concentration for the Mutagenicity Assay.

8.9.2 Maintain at least five different analyzable concentrations of the test item with half log intervals between test points (3.16) for DRF experiment. Smaller intervals (1.5 to 3) may be appropriate when a concentration-response is being investigated or as decided upon by Study Director in conjunction with Sponsor.

8.9.3 Conduct the experiment using one or two tester strains (e.g. TA100 and / or TA98) by Plate Incorporation method (refer section 8.11 for treatment) in with and without metabolic activation system or as per the study plan.

8.10 Mutagenicity Assay

8.10.1 Select the doses based on the results of DRF. If the test item is soluble and non-cytotoxic, select 5 mg/plate or 5 µl/plate as the top concentration. For precipitating, non-cytotoxic test items, select the lowest precipitating concentration as the top concentration. Test items that are cytotoxic below 5 mg/plate should be tested up to a cytotoxic concentration as high concentration.

8.10.2 Maintain spacing intervals ranging from approximately 2 to 3.16 among selected concentrations for mutagenicity assay.

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8.10.3 Perform mutagenicity assay using five tester strains viz Salmonella typhimurium TA1537, TA1535, TA98, TA100 and TA102 or *E. coli* or as specified in Study Plan.

8.10.4 Conduct Mutagenicity assessment using direct plate incorporation or pre incubation treatment method or as specified in Study Plan.

8.10.5 Confirmatory assay shall be performed on need basis to confirm the negative or equivocal results obtained in the initial mutagenicity assay. Confirmatory assay shall be performed with modified experimental conditions such as treatment method, metabolic activation conditions, concentration spacing of test item treatment etc. or as specified in Study Plan.

8.11 Direct Plate Incorporation Method

8.11.1 This method includes the addition of test item/positive control/vehicle control, S9 mixture or phosphate buffer and bacterial tester strain to the top agar followed by direct pouring of the contents onto the minimal glucose agar (MGA) plate.

8.11.2 Ensure that sufficient number of sterile MGA plates are available for treatment.

8.11.3 Use duplicate/triplicates plates per concentration of test item / positive and vehicle controls or as specified in the study plan.

8.11.4 Arrange the sterile containers (tubes/vials) in to LAFC/BSC.

8.11.5 Add 0.1 mL of test item or vehicle or positive control, 0.5 mL of S9 mixture or Phosphate buffer (PB) and 0.1 mL of tester strain in to sterile containers (tubes/vials). Then add 1.6 mL for mini AMES test and 2.0 mL of top agar (maintained approximately at 45-47°C) for BRMT supplemented with histidine-biotin solution for Salmonella strains and tryptophan for *E.Coli* strain.

8.11.6 Mix the added contents and pour on to MGA plates. Use vortexer to mix the contents if required.

8.11.7 Rotate plates gently to distribute top agar uniformly across the minimal agar plates. Take care not to spill the contents and close the plates with lids.

8.11.8 Keep the plates in still position until top agar solidifies. Then transfer the plates to incubator for incubation at 37 ± 2˚C for 48 to 72 hours. Preferably invert the plates for incubation. Document the treatment details in **FORM-GLP-TOX-0075.**

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8.12 Pre-incubation Method

8.12.1 In this method, mixture of test item/positive control/ vehicle control, S9 mixture or phosphate buffer and bacterial tester strain are incubated for 20-30 min prior to plating. This enables test item exposure at higher concentrations attributing to increased sensitivity.

8.12.2 Use duplicate/triplicates plates per concentration of test item / positive and vehicle controls or as specified in the study plan.

8.12.3 Arrange the sterile containers (tubes/vials) in to LAFC/BSC.

8.12.4 Add 0.1 mL of test item or vehicle or positive control, 0.5 mL of S9 mixture or Phosphate buffer (PB) and 0.1 mL of tester strain in to sterile containers (tubes/vials).

8.12.5 Incubate the containers (tubes/vials) covered with aluminum foil or caps preferably in a shaking water bath/ Incubator approximately at 75 - 100 rpm (if shaking is required) and at 37 ± 2ºC for 20-30 minutes or as specified in the Study Plan.

8.12.6 Post incubation, transfer the tubes from water bath to LAFC/BSC and add 1.6 mL for mini AMES and 2.0 mL of top agar for BRMT (top agar maintained approximately at 45-47˚C) supplemented with histidine-Biotin solution for Salmonella strains and tryptophan for *E.coli* strain.

8.12.7 Mix the added contents and pour on to MGA plates. Use vortexer to mix the contents if required.

8.12.8 Rotate plates gently to distribute top agar uniformly across the minimal agar plates. Take care not to spill the contents and close the plates with lids.

8.12.9 Keep the plates in still position until top agar solidifies. Then transfer the plates to incubator for incubation at 37 ± 2˚C for 48 to 72 hours. Preferably invert the plates for incubation.

8.12.10 Document the details using format **FORM-GLP-TOX-0075.**

Note: *Addition of top agar (1 or 3mL) does not have any impact on the outcome of the study.*

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8.13 Control Plates

8.13.1 Maintain vehicle control and positive control plates with each experiment for all the tester strains. Treat vehicle and positive controls under the same treatment conditions as the test item in duplicate/triplicates.

8.13.2 Maintain untreated and negative control plates where other than routine solvent is used, or as and when required.

8.14 Genotype Confirmation of Tester stains

8.14.1 Periodic Confirmation of genotype markers of tester strains will be performed as per sop **SOP-GLP-TOX-0046**.

8.14.2 Perform genotype confirmation of tester strains along with mutagenicity assay with minimum of any three selective markers preferably R-factor, rfa, auxotroph mutation and UV sensitivity or as per study plan.

8.14.3 Perform the genotype of selected markers in Single plate or divide plate into quadrants for each strain as applicable.

8.14.4 Streak the nutrient agar plate/MGA plate with tester strain culture and allow to dry inside the BSC/LAFC.

8.14.5 Place filter paper discs on quadrants labelled for antibiotic and crystal violet (CV) and add approximately 10 µL of respective solutions on to designated discs.

8.14.6 For UV sensitivity, expose one half of the plate to UV light as defined in the procedure stated in **SOP-GLP-TOX-0046.**

8.14.7 Incubate the plates at 37 ± 2˚C for 24 to 48 hours.

8.14.8 After 24 to 48 hours of incubation, take plates out from the incubator (not mandatory to take along with study treated plates).

8.14.9 Observe genotype plates for zone of inhibition or growth and enter the details of genotype confirmation in form **FORM-GLP-TOX-0075**.

8.15 Observations and Colony Counting

8.15.1 Observe the treated plates with naked eye or under microscope (4 to 10x magnification) for background lawn (cytotoxicity) and precipitation (interference

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with background lawn) and document the details in format **FORM-GLP-TOX-0072**. Compare with respective strain control plates.

8.15.2 Record the precipitation details as listed in the table below;

Description of Precipitation Observation during colony counting
No Precipitation – NP
Non-Interfering precipitation – NIP
Interfering precipitation – IP

8.15.3 Record the background lawn details as listed in the table below;

Description of Background lawn Observation during colony counting
No thinning to mild thinning - N
Moderate thinning – T Extreme thinning to absence of lawn - A

8.15.4 Count the number of revertant colonies manually or using a colony counter and document using format **FORM-GLP-TOX-0072**. For positive control plates or other treatment plates with colony counts with more than three hundred, divide the plates into equal quadrants. For numerous colonies on plates, count any one quadrant and multiply with appropriate factor to get the final count, provided the colonies size is uniform and are evenly distributed over the entire surface of the plate.

8.15.5 If microscope is used, refer **SOP-GLP-TOX-0084** and document details of usage log.

8.15.6 Refer **WI-GLP-TOX-0027** for pictorial description of revertant colonies.

8.15.7 If revertant colony counting is not completed in the same day of incubation completion, then store the plates under refrigerator (2-10°C) conditions and complete colony counting with in three business working days.

8.16 Acceptance Criteria: Assay is considered acceptable if it meets the following criteria;

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8.16.1 Untreated/ Vehicle Control: Number of revertant colonies should be within 3 standard deviations (3SD) of the historical vehicle/untreated control range.

8.16.2 Positive Control: Positive control plates should show a minimum of three-fold increase in the number of revertant colonies when compared to the vehicle control plate.

Note: *Compare the revertant colonies of PCs with concurrent vehicle control or untreated control revertant colonies maintained for the experiment.*

8.16.3 Test Item: The highest selected concentration shall exhibit evident toxicity or shall exhibit limited solubility as demonstrated by the preliminary dose range-finding study or shall extend to recommended highest test concentration by regulatory guidelines (OECD 471). Exceptions to the above mentioned are scientifically justified.

8.16.4 Not more than 5% of the plates are lost through contamination or some other unforeseen event.

8.16.5 ***Note:*** *In the situation where more than 5% of the plates were lost, then consider those plates where contamination is not interfering with colony counts. In situation where all plates pertaining to any one of the strains were lost due to contamination, then consider this as strain contamination and exclude those strain plates for counting without invalidating the study and re-perform the treatment with excluded strains.*

8.16.6 Enter the details of contamination if any in format **FORM-GLP-TOX-0072.**

8.17 Data Compilation

8.17.1 Data shall be presented in terms of number of revertant colonies grown per plate.

8.17.2 Calculate Mean and SD for revertant colonies from each test item concentration and controls.

8.17.3 Calculate fold increase in revertant colonies for test item concentrations and positive controls in comparison to respective vehicle controls.

8.18 Interpretation of Results

8.18.1 Consider test item as mutagenic or positive if;

- The number of revertant colonies in TA100, TA102 and *E.coli* exhibit more than two fold increase when compared to respective vehicle control.

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- The number of revertant colonies in TA1535, TA1537 and TA98 exhibit more than threefold increase when compared to respective vehicle control.
- There is a concentration related increase over the range tested and/or a reproducible increase at one or more concentrations in the number of revertant colonies in at least one tester strain with or without metabolic activation.

8.18.2 Biological relevance of the results shall be considered first. Statistical tool may be used for evaluating the test results on need basis.

8.18.3 There is no need for verification of clear positive response.

8.18.4 Equivocal results shall be clarified preferably by further testing using modification of experimental conditions (concentration spacing, method of treatment and metabolic activation conditions) if necessary. Results may remain equivocal or questionable regardless the number of the times the experiment is repeated. The final discretion of the interpretation will lie with the Study Director.

8.19 Historical Data

8.19.1 Update in house historical control data with data from valid experiments only. Determine values not limited to Mean, SD, 2SD, minimum value, maximum value and number of plates (n). Use Microsoft Excel to calculate these parameters and compile in-house historical control data. Periodically (at least once in year) take a hard copy print of the compiled in-house historical control data and maintain as facility record for reference and archive as per **SOP-GLP-AC-0001.**

8.20 Study Report

8.20.1 Prepare the study report as per procedure stated in **SOP-GLP-GEN-0023**. Include details not limited to that of bacterial tester strains (characteristics etc.), test conditions and results including cytotoxicity, precipitation, individual plate counts, mean number of revertant colonies and standard deviation, concurrent controls data and historical data with e.g. fold increase, mean and standard deviations.

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9.0 REFERENCE(S):

SOP Number	Title of the SOP or Reference Description
NA	OECD guideline for testing of chemicals (No. 471, Corrected 26 June 2020, titled Bacterial Reverse Mutation Test)
NA	Toxicological Principles for the Safety Assessment of Food Ingredients Redbook 2000 (No. IV.C.1.a. Bacterial Reverse Mutation Test)
NA	Revised methods for the Salmonella mutagenicity test. Dorothy M. Maron and Bruce N. Ames; 1983; Mutation Research; 113, pages: 173-215.
NA	The Ames Salmonella/microsome mutagenicity assay. Kristien Mortelmans, Errol Zeiger; 2000; Mutation Research; 455, pages: 29–60.
NA	The bacterial tryptophan reverse mutation assay with Escherichia coli WP2. Kristien Mortelmans, Edward S. Riccio; 2000; Mutation Research; 455, pages: 61-69.
SOP-GLP-GEN-0023	Preparation of Study Report & Report Amendment
SOP-GLP-GEN-0021	Operation, Validation, Cleaning and Maintenance of Laminar Air Flow and Biosafety Cabinets
SOP-GLP-TOX-0045	Preparation of Reagents and Media for Bacterial Reverse Mutation Test (Ames test)
SOP-GLP-TOX-0046	Methods of Storage and Revival of Tester Strains for Bacterial Reverse Mutation Assay (Ames Test)
SOP-GLP-TOX-0084	Operation, Cleaning and Maintenance of Microscopes
SOP-GLP-TOX-0044	Positive Controls for Genotoxicity Studies: Receipt, Labeling, Handling, Storage and Disposal
SOP-GLP-TC-0003	Handling, Storage, Distribution, and Return of Test and

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	Reference Items Used for Non clinical safety studies.
SOP-GLP-SHE-0004	Bio Waste Management
SOP-GLP-SHE-0002	Chemical & Biological Spillages and Cleaning Procedures
SOP-GLP-SHE-0003	Hazardous Waste Management
SOP-GLP-AC-0001	Receipt of Records, Materials to Archives, Procedures for archiving, allotting accession number, indexing and assigning retention period.

10.0 ANNEXURE(S):

Annexure No.	Details/Title of the Annexure
WI-GLP-TOX-0027	Pictorial Description of Revertant Colonies
WI-GLP-TOX-0029	List of Positive Controls (BRMT)
FORM-GLP-TOX-0067	Bacterial Culture Inoculation and Density Measurement
FORM-GLP-TOX-0069	Test Item Formulations Preparation Record
FORM-GLP-TOX-0070	Positive Control Formulations Preparation Record
FORM-GLP-TOX-0071	S9 Mixture Preparation Record
FORM-GLP-TOX-0072	Observation of Revertant Colonies
FORM-GLP-TOX-0075	Treatment Record-BRMT Assay
FORM-GLP-TOX-0076	S9 Receipt and Utilization Record
FORM-GLP-TOX-0077	Bacterial Reverse Mutation Test-Solubility and Precipitation Test

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11.0 CHANGE HISTORY

Quality Record No	Brief details of changes
CC1000088093	• SOP-GLP-TOX-0048 is revised to address the CAPA (No: CAPA/SE/21-029) and to refine the procedures and formats for more clarity and also for better representation of the assay conduct. • Storage conditions of treatment plates after completion of incubation period is provided. • Addition of genotype characterization during conduction of study is newly included in the procedures • Description of Precipitation and background lawn of colonies were modified. • Consideration and review of contamination of treated plates in the study is defined. • SOP, Formats and Work Instructions templates are updated in line with current DOCHUB template. • FORM-GLP-TOX-0067-Revised • FORM-GLP-TOX-0069-Revised • FORM-GLP-TOX-0070-Revised • FORM-GLP-TOX-0071-Revised • FORM-GLP-TOX-0072-Revised • FORM-GLP-TOX-0073-Withdrawn • FORM-GLP-TOX-0074-Merged into FORM-GLP-TOX-0075 • FORM-GLP-TOX-0075-Revised • FORM-GLP-TOX-0076-Revised • FORM-GLP-TOX-0077-Revised

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Intention	Employee ID	User Name	Date (IST) DD/MM/YYYY
Prepared By	P00075663	Tulasi Sukanya Kotha.	21/07/2022 15:32:30
Peer Reviewed By	P00073722	Srinivasa Rao Boddapati.	21/07/2022 15:36:44
Peer Reviewed By	P00078225	Parvathi M V S.	21/07/2022 15:37:50
QA Reviewed By	P00070211	Nishant Christian.	21/07/2022 17:45:33
Approved By Test Facility Management	P00068654	D Ravi Krishna Babu.	21/07/2022 19:03:22

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MINI AMES ASSAY WITH HISTIDINE REQUIRING TESTER STRAINS OF *SALMONELLA TYPHIMURIUM*

Study Number : **XX**

GLP status : Non-GLP

Sponsor : xxxxxxxxxxxxxxxxxxxx

Test Facility : Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad - 500 049, Telangana, India.

1. STUDY OBJECTIVE

The objective of the study is to assess the mutagenic potential of test item using histidine requiring tester strains of *Salmonella typhimurium*.

2. TEST ITEM

Test item: xxxxxxxxx
Batch no: xxxxxxxxx

3. TEST SYSTEM

The study will be carried out using two different tester strains of *Salmonella typhimurium* bacteria. The details of the strain are summarised in the below table:

Bacteria	Tester Strain	Reversion Event
S. typhimurium	TA98	frame-shift
S. typhimurium	TA100	base-pair substitution

All the tester strains were originally obtained from Molecular Toxicology, INC., Boone, NC, USA.

4. POSITIVE CONTROLS

The details of the positive control items to be used are given in the following table:

Chemical name & CAS No.	Concentration (µg/plate)	Solvent	Strain(s)	S9
2-Aminoanthracene (2-AA) (613-13-8)	5	DMSO	TA 98 TA 100	+
2-Nitrofluorene (607-57-8)	5	DMSO	TA98	–
Sodium azide (26628-22-8)	5	DMSO	TA100	–

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5. MEDIA & SOLUTIONS

All media and reagents required for the study will be prepared and documentated as per the applicable test facility SOPs.

6. SOLUBILITY AND PRECIPITATION TEST

Test item solubility will be checked in suitable solvent. Test item formulations will be prepared freshly on the day of treatment, prior to exposure to the test system. Test compound will be assessed for precipitation check in the solvent and on the agar plate. Primary stock of test compound yielding highest concentration will be prepared using vehicle. 100 µL of test compound formulation will be taken and added to top agar to achieve highest desired test concentration. The tube will be vortexed gently and plated on to minimal agar plate. If visible precipitation is observed on the plate, the procedure will be repeated with lower dilutions of the stock solutions to achieve non interfering precipitation. The precipitation will be graded and decision on the highest test concentration for testing will be taken based on the results of precipitation test.

7. DOSE FORMULATION

Required quantity of test compound will be weighed and transferred to the appropriate size container. Required volume of vehicle will be added to it to make the highest desired concentration and vortexed/sonicated until clear solution is attained. Lower strengths of test compound will be prepared by serial dilution of highest strength using vehicle.

8. STUDY DESIGN

Test compounds will be tested at a minimum of five concentrations employing Pre-incubation method (With and without S9).

Test Concentration (µg /well)	Treatment Method	Metabolic Activation
#	Pre-Incubation	Present (10 % v/v S9 Cofactor mix)
		Absent (Phosphate Buffer)

Test concentrations may be different based on the Solubility/Precipitation test results.

8.1 Revival of Tester Strains

Frozen permanents of bacteria (inoculums) will be thawed and cultures will be inoculated using flask containing Oxoid Nutrient Broth (ONB) for (~12 to 16 hours)

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and incubated overnight in an incubator equipped with a shaker at 37±2°C with shaking at 110 ± 10 revolutions per minute. These overnight grown cultures containing 10^8 - 10^{10} CFU/mL will be used for the assay conduct.

8.2 Pre-incubation Method

For pre-incubation method, a pre-incubation step is included in which, vehicle control or test item or positive control (100 µL), S9 mix/phosphate buffer (0.5 mL) and bacterial culture (100 µL) will be mixed and incubated for minimum period of 20 - 30 minutes at 37 ± 2°C in a shaking water bath at 75-100 rpm. After incubation, 2 mL of molten agar with histidine-biotin solution maintained at 45 ± 3°C will be added to the mixture followed by rapid vortexing and pouring on to pre-labelled MGA plates. When the agar is set, the plates will be inverted and incubated. The plates will be suitably labelled to clearly identify the name of the tester strain, experiment method, test item concentration and vehicle control group.

Note: Untreated controls will be maintained in case of new vehicle usage.

8.3 Incubation

All treated plates will be incubated at 37± 2°C at ambient air for 48 to 72 hours.

9. EVALUATION OF RESULTS

9.1 Cytotoxicity and Precipitation

Cytotoxicity will be assessed in terms of diminution of background bacterial lawn, presence of microcolonies and/or reduction in the number of revertant colonies. Precipitation will be graded on the plate, based on the amount of visible precipitate on the plate.

Precipitation Observation	
NP	No Precipitation
NIP	Non-Interfering precipitation
IP	Interfering precipitation
Background Lawn Observation	
N	No thinning to mild thinning
T	Moderate thinning
A	Extreme thinning to absence of lawn
Note:	Revertant colonies/micro colonies will not be counted where test item shows absence or extreme reduction of lawn.

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9.2 Colony Counting

Revertant colonies will be counted manually after incubation of 48-72 hours.

10. ANALYSIS OF RESULTS

Individual plate counts will be recorded separately and the mean and standard deviation for each treatment/concentration will be determined. Revertant colonies obtained in test items and positive controls treated plates will be compared to respective vehicle control and fold increase will be calculated for each test concentration.

10.1 Acceptance Criteria

The test will be considered valid if all the following criteria are met:

- The negative control counts should be comparable with the in-house data and published literature.
- The positive control items induce clear increases (minimum of 2-fold in comparison with concurrent vehicle controls) in the number of revertant colonies, confirming discrimination between different strains, and an active S9 preparation.
- Not more than 5% of the plates are lost through contamination or some other unforeseen event.

10.2 Evaluation Criteria

The test compounds shall be considered positive if

- Concentration related evident increase over the range tested in the number of revertant colonies per plate in at least one tester strain.
- The increase is considered significant if the total number of revertants is greater than two fold to that of the revertants in their respective controls.

Results which only partially satisfy the above criteria will be handled on a case-by-case basis. Biological relevance will be taken into account, for example consistency of response within and between concentrations. Additional confirmatory experiment shall be conducted if deemed necessary. If the above criteria are not met, the test compounds will be considered as negative.

11. REPORTING

11.1. Draft report

A draft report containing all procedures and results will be issued for study Sponsor's review.

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11.2. Final report

Study report will be finalized after the review comments have been addressed appropriately and an agreement is obtained from Sponsor. After reciprocal agreement, one original hard copy of final study report will be generated. One electronic copy of the final report will be sent to the study Sponsor.

The time span of receiving Sponsor's comments shall not exceed two months. If no comments/response are received from the Sponsor within/end of two months period, the SD shall contact the Sponsor and ask for the reviewed draft report. If the SD does not get any response from the Sponsor within two weeks from his/her reminder mail/letter, the draft report shall be archived without finalization. The report shall be finalized once the Sponsor gives their review comments and after obtaining a formal approval from the Sponsor.

12. STORAGE/RETENTION OF DATA

Raw data and other documents generated as a result of this study will be retained at the Non-GLP archives of APSL for a minimum period of 9 years from the date of study completion. After completion of 9 years, the study Sponsor will be contacted for continuation of archival storage at the Test Facility or to have the materials returned to them.

13. TEST GUIDELINES AND REFERENCES

The following guideline/publications are used as reference documents for designing the study;

a. Organization for Economic Co-operation and Development (OECD): Guidelines for Testing of Chemicals: *In Vitro* Mammalian Cell Micronucleus Test 487: Adopted: 29 July 2016.

b. Flamand N, et al. (2001): Mini mutagenicity test: a miniaturized version of the Ames test used in a prescreening assay for point mutagenesis assessment. Toxicology, In Vitro. 15: 105-114.

c. Maron, D.M. and Ames, B.N. (1983): Revised methods for the Salmonella mutagenicity test. Mutat. Res., 113: 173–215.

14. SPONSOR APPROVAL

Sponsor Contact detials	Apporved Signatrures
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx	

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